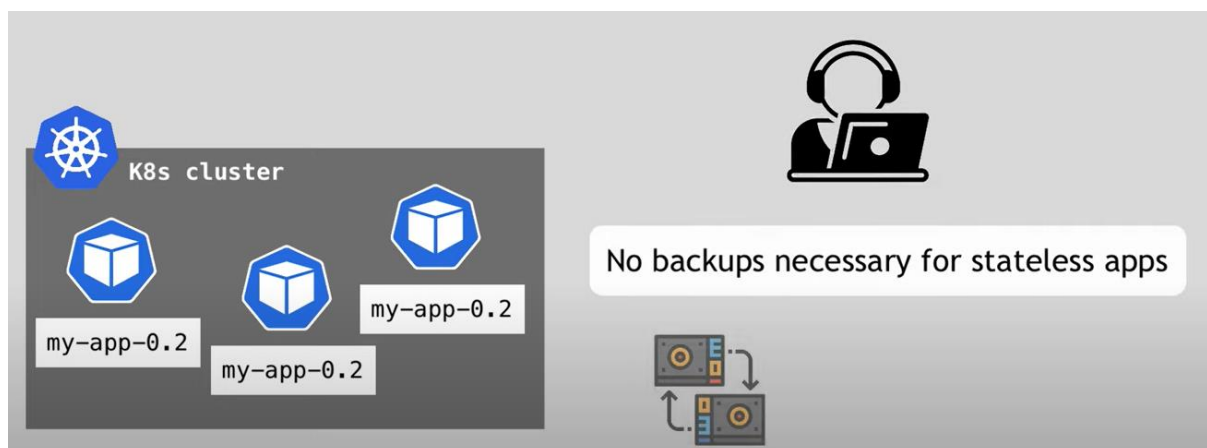
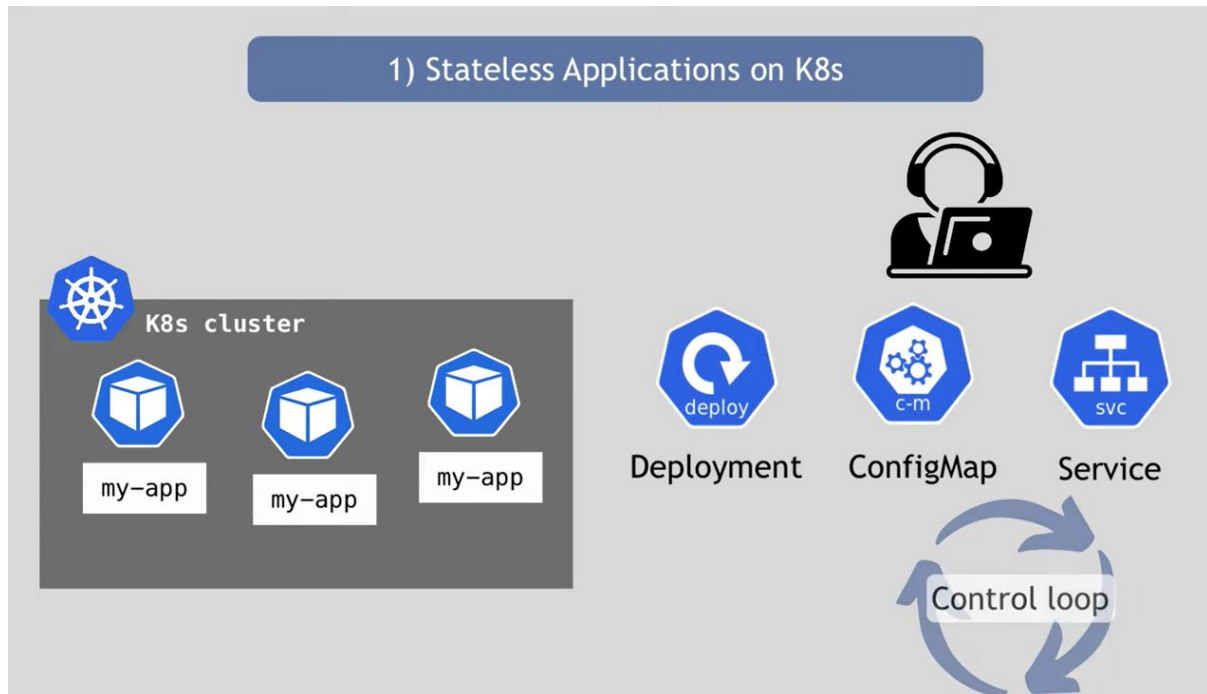


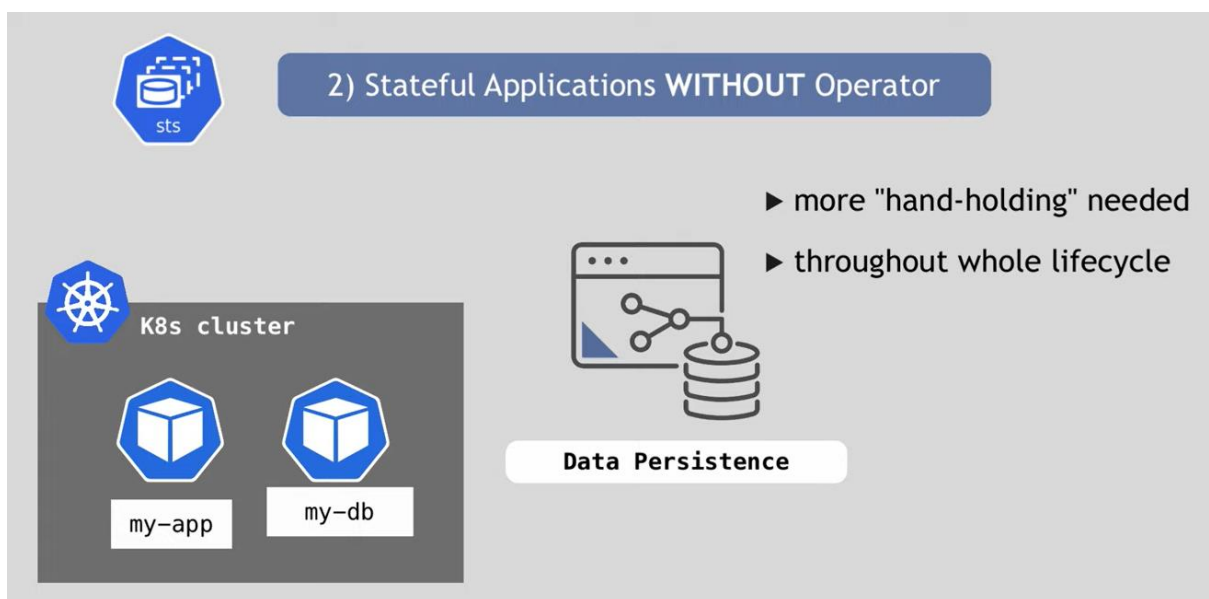
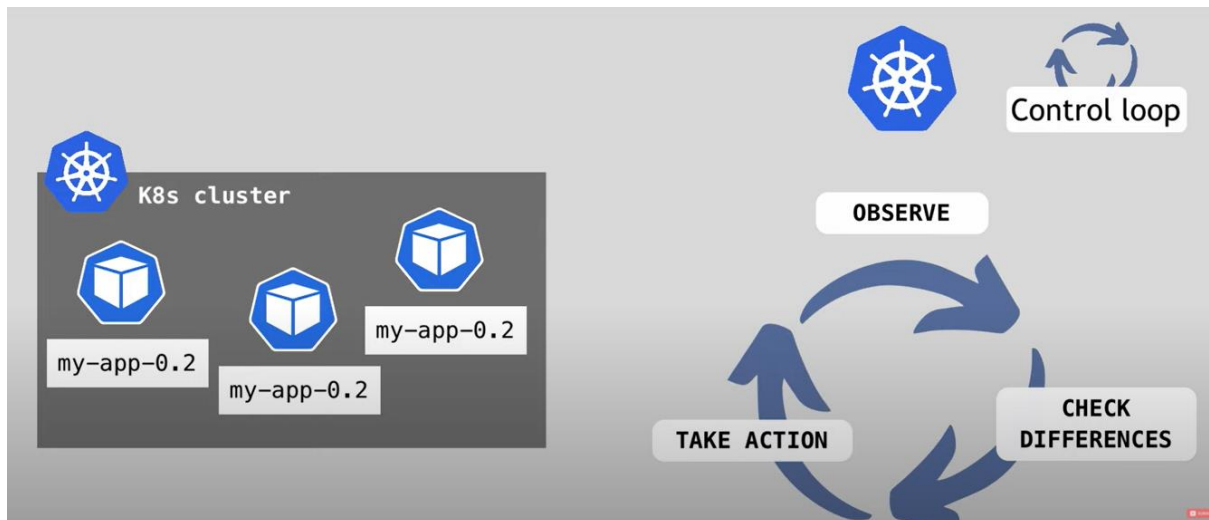
## Why K8s Operators?

Used for Stateful applications

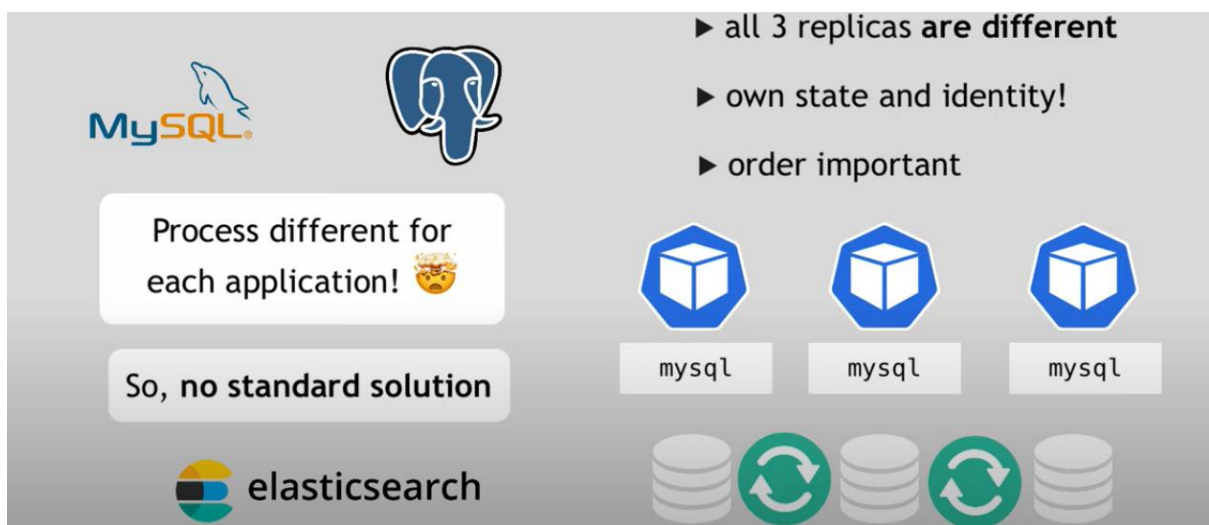
UC# Stateful application without the Operator



No control necessary after app deployed.



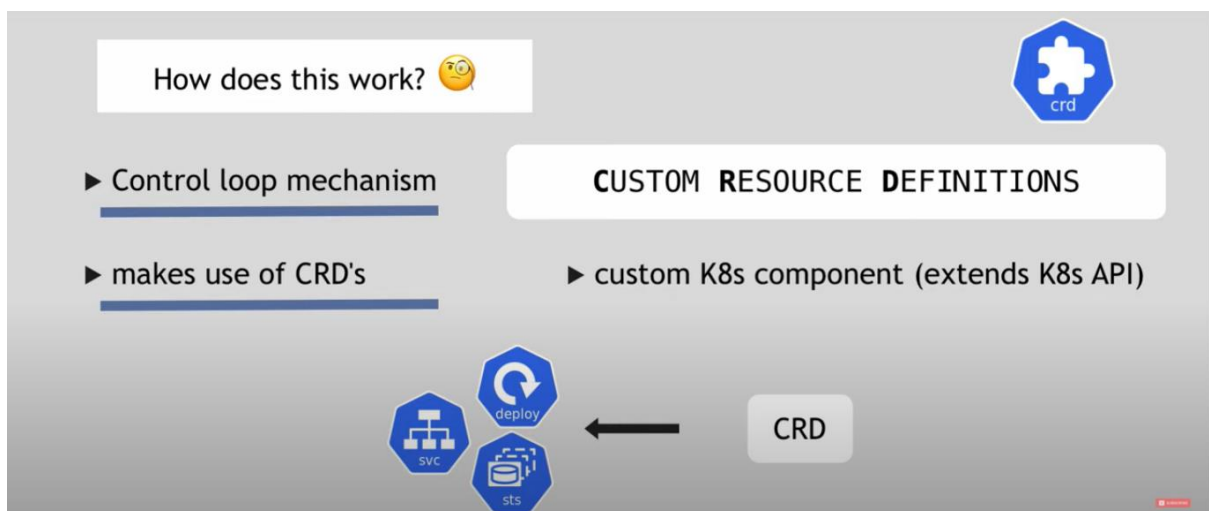
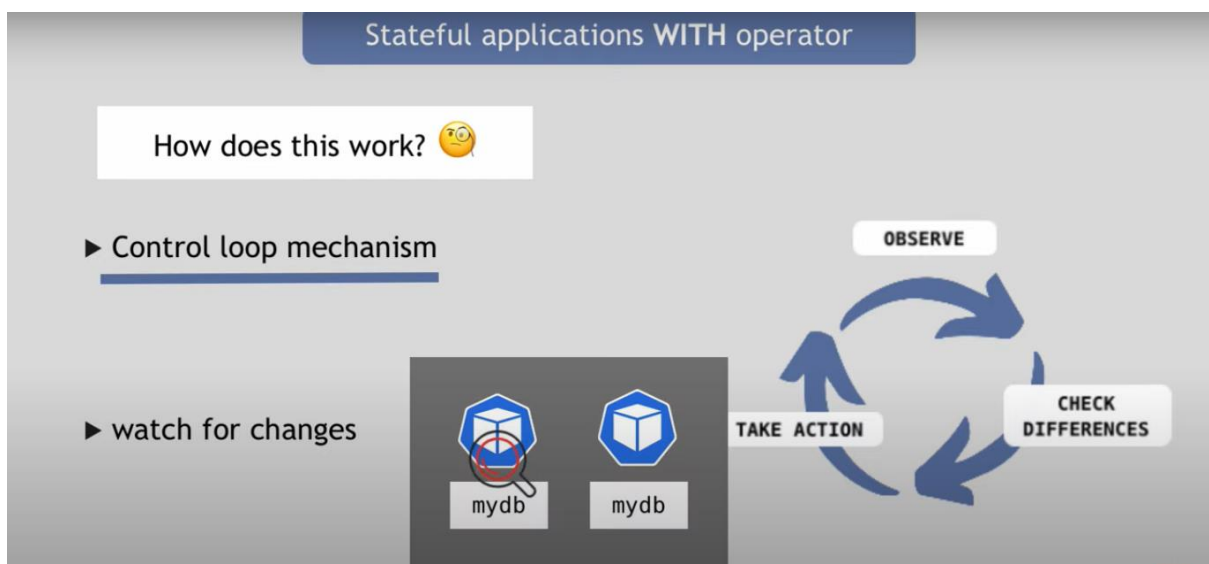
If we create three different replicas of mysql, all three are different, each have their own state and Identity, means order is important.

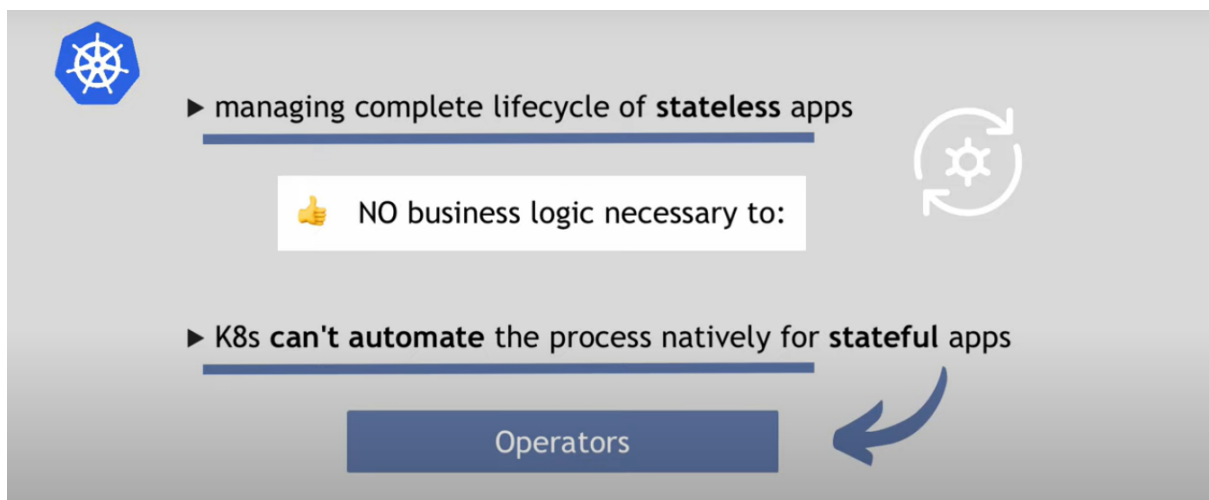
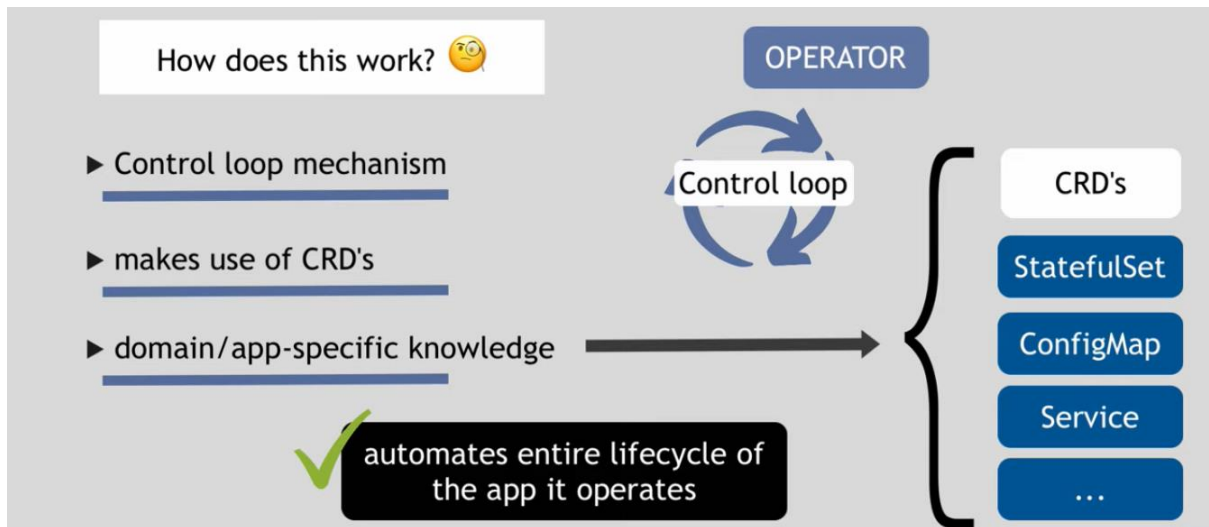


Stateful application required manual intervention necessary.

# What is Operator (custom control loop in K8s)

Replaces human operator with S/W operators





The diagram illustrates a MySQL cluster architecture. On the left, a group of three people icons is associated with the MySQL logo and a blue button labeled "OPERATOR". To the right, four MySQL nodes are shown, each represented by a blue cube icon and a white box labeled "mysql". Below the operator icon, a lightbulb icon is followed by a list of tasks:

- How to create the mysql cluster
- How to run it
- How to synchronize the data
- How to update

The diagram displays four Kubernetes operators, each with a group of people icon and a label in a teal box:

- prometheus-operator**: Associated with the Prometheus logo (a red circle with a white flame).
- postgres-operator**: Associated with the PostgreSQL logo (a blue elephant).
- elastic-operator**: Associated with the Elasticsearch logo (a yellow and red circle).
- mysql-operator**: Associated with the MySQL logo.

Operatorhub